Derivatives and the Chain Rule

Mathematics Foundations — Session 3

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Today's Goal

- Read a derivative as an instantaneous rate of change, straight off a graph
- Differentiate polynomials, exponentials, and logarithms
- Apply the chain rule to a composed function, one layer at a time
- See why the chain rule is backpropagation

→ **You should be able to:** differentiate a composed function by hand and connect it to backprop in one sentence

Plan

1. Opening Quiz
2. The Derivative
3. Differentiation Rules and Properties
4. The Chain Rule
5. Backpropagation
6. Exercises
7. Recap
8. To Do for Next Session

Opening Quiz

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Opening Quiz: Matrix Multiplication & the Forward Pass

Q1.	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = ?$	A) $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$	B) $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$	C) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
Q2.	A is 2×3, B is 3×4: size of AB?	A) 3×3	B) 2×4	C) undefined
Q3.	For all square matrices, $AB = BA$?	A) True	B) False	
Q4.	$\det \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = ?$	A) 2	B) 10	C) -2
Q5.	Layer $W = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, input $x = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$: $h = Wx = ?$	A) $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$	B) $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$	C) $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Opening Quiz: Matrix Multiplication & the Forward Pass

Q1.	$\left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)\left(\begin{smallmatrix}1\\1\end{smallmatrix}\right) = ?$	A) $\left(\begin{smallmatrix}3\\7\end{smallmatrix}\right)$	B) $\left(\begin{smallmatrix}4\\6\end{smallmatrix}\right)$	C) $\left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$
Q2.	A is 2×3, B is 3×4: size of AB?	A) 3×3	B) 2×4	C) undefined
Q3.	For all square matrices, $AB = BA$?	A) True	B) False	
Q4.	$\det\left(\begin{smallmatrix}2&1\\4&3\end{smallmatrix}\right) = ?$	A) 2	B) 10	C) -2
Q5.	Layer $W = \left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$, input $x = \left(\begin{smallmatrix}2\\1\end{smallmatrix}\right)$: $h = Wx = ?$	A) $\left(\begin{smallmatrix}2\\3\end{smallmatrix}\right)$	B) $\left(\begin{smallmatrix}3\\1\end{smallmatrix}\right)$	C) $\left(\begin{smallmatrix}2\\1\end{smallmatrix}\right)$

Answers

Q1 A: $(1+2,\ 3+4)$ **Q2** B: $(2\times \mathbf{3})(\mathbf{3}\times 4) = 2\times 4$ **Q3** B: not commutative **Q4** A: $2\times 3 - 1\times 4 = 2$ **Q5** A: $(2,\ 2+1)$

The Derivative

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Introduction

In Sessions 1 and 2, we went into **vector spaces** and **linear transformations** $\mathbb{R}^n \to \mathbb{R}^m$.

For this class, we take a detour in $\mathbb{R}$ and study functions $\mathbb{R} \to \mathbb{R}$.

This is called **Analysis** (as opposed to **Algebra**).

Recall: What Is a Function?

Definition

A **function** is a mapping from a space to another, $f : X \to Y$. To each element of X, it gives a **unique** element of Y.

Examples:

- $f(x) = x$ (identity)
- $f(x) = 0$ (constant)
- $f(x) = x^2$: what is the X space? What is the Y space?
  $\to X = \mathbb{R}$ (any x works), and the values are in $Y = [0, +\infty)$ (we can also say $Y = \mathbb{R}$: then not every y is reached).

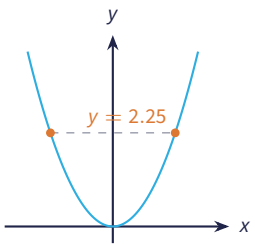
Vocabulary

A function $f : X \to Y$ is said to be:

- **Injective**: two different inputs always give two different outputs:
  $f(a) = f(b) \Rightarrow a = b$
- **Surjective**: every $y \in Y$ is reached: for all $y \in Y$, there is an $x \in X$ with $f(x) = y$
- **Bijective**: both injective and surjective: every $y \in Y$ is reached **exactly once**

△**!** The definitions depend on X and Y.

Vocabulary: Example



$f(x) = x^2$ is:

- **injective** if $X = [0, +\infty)$
  (on $\mathbb{R}$, $f(-1.5) = f(1.5)$: not injective)
- **surjective** if $Y = [0, +\infty)$
  (on $Y = \mathbb{R}$, $y = -1$ is never reached)
- **bijective** if $f : [0, +\infty) \to [0, +\infty)$

Function Composition

Let $x \in \mathbb{R}$:

- $f \circ g$ is: $(f \circ g)(x) = f(g(x))$ (first g, then f)
- $f + g$ is: $(f + g)(x) = f(x) + g(x)$
- $f \times g$ is: $(f \times g)(x) = f(x) \times g(x)$

$\triangle$**!** $f \circ g \neq f \times g$

For $g : X_g \to Y_g$ and $f : X_f \to Y_f$, $f \circ g$ exists if $Y_g \subset X_f$: the outputs of g must be valid inputs of f.

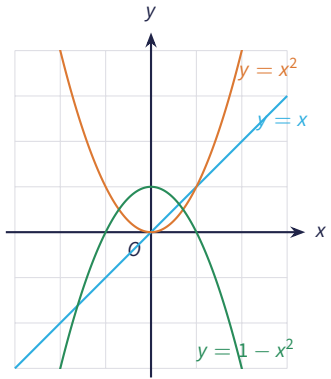
Function Composition: Example

Example: $f(x) = x^2, \qquad g(x) = x + 1$

- $f \circ g$ is: $f(g(x)) = (x+1)^2$
- $g \circ f$ is: $g(f(x)) = x^2 + 1$
- $g \times f$ is: $(x+1)\,x^2 = x^3 + x^2$
- $f \times g$ is: $x^2(x+1) = x^3 + x^2$ (same: $\times$ is commutative)

$\Rightarrow f \circ g \neq g \circ f$: the order matters for composition.

Plotting a Curve



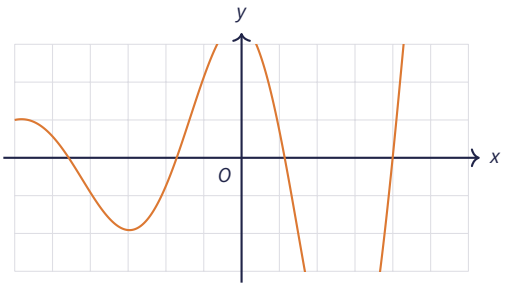
Plot, point by point:

- $y = x$
- $y = x^2$
- $y = 1 - x^2$

e.g. for $y = x^2$: $(0, 0)$, $(\pm 1, 1)$, $(\pm 2, 4)$.

Remark: a function is defined by its curve :)

Rate of Change of a Curve



Open questions:

- Can we know if f increases or decreases?
- Can we know if f plateaus?
- Can we define the “speed”, the rate of change of f?

Can we have **numbers** to represent all these?

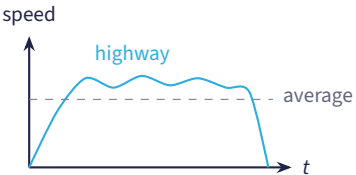
Definition of Speed (1/2)

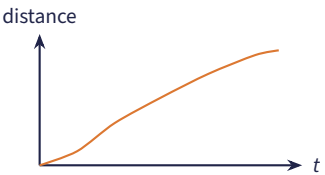
Driving from Marseille to Paris (about 750 km) in 7 h 30: what is the average speed?

$$v = \frac{\Delta d}{\Delta t} = \frac{750\ \text{km}}{7.5\ \text{h}} = 100\ \text{km/h}$$

Δ denotes a difference: $\Delta d = d_{\text{end}} - d_{\text{start}}$

But this is only the **average**:

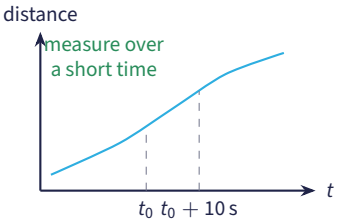




Definition of Speed (2/2)

So $v = \dfrac{\Delta d}{\Delta t}$ is only the average. Can we guess what the driver sees at a **particular time** t_0?

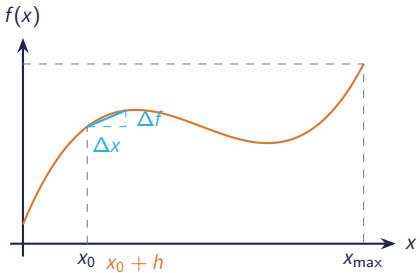




The speedometer measures the distance travelled over a **very short** time:

$$v(t_0) \approx \frac{d(t_0 + 10\,\mathrm{s}) - d(t_0)}{10\,\mathrm{s}}.$$

Getting Back to Functions



The **average** rate of change on $[0, x_{\max}]$ is $\dfrac{f(x_{\max}) - f(0)}{x_{\max}}$.

What is the "speed" at x_0?

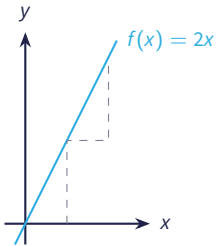
Take a small step h:

$$\frac{\Delta f}{\Delta x} = \frac{f(x_0 + h) - f(x_0)}{h}$$

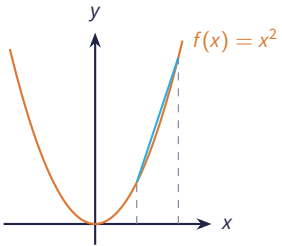
Notation: Δ is a big difference, d is a small one: speed $= \boxed{\dfrac{df}{dx}}$

Getting Back to Functions: Example

Example: $h = 1, x_0 = 1$, rate of change $\dfrac{f(x_0 + h) - f(x_0)}{h}$:



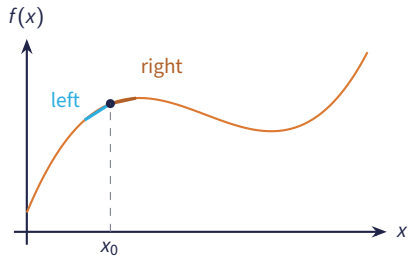
$f(x) = 2x$: $\dfrac{2 \times 2 - 2 \times 1}{1} = 2$



$f(x) = x^2$: $\dfrac{2^2 - 1^2}{1} = 3$

For x^2, smaller h: $h = 0.1 \to 2.1$, $h = 0.01 \to 2.01$: the rate gets closer to 2.

One Remark

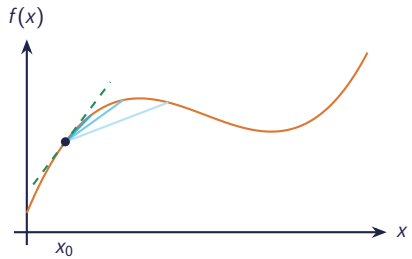


Right (step to the right):
$$\frac{f(x_0 + h) - f(x_0)}{h}$$

Left (step to the left): $\dfrac{f(x_0) - f(x_0 - h)}{h}$

Different (almost equivalent) ways of computing the speed: for small h, they give almost the same number.

What Is the Effect of the Size of h?



- The **larger** h, the **less precise** it is.
- The **smaller** h, the **more instantaneous** it is.

When $h \to 0$, it is called the **instantaneous velocity** / **rate of change**.

$\Rightarrow$ This is the **derivative** of f at x_0 (definition below).

Dashed: the tangent, the limit of the secants.

The Derivative as an Instantaneous Rate of Change

The derivative

A function is said to be **derivable / differentiable** at a point x_0 if the following limit **exists and is finite**:

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

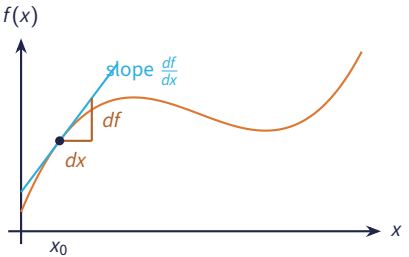
How to go from one to the other? Set $h = x - x_0$: then $x = x_0 + h$, and $x \to x_0$ means $h \to 0$.

The Derivative as an Instantaneous Rate of Change

Notation: it is noted

$$f'(x_0) \qquad \text{or} \qquad \left.\frac{df}{dx}\right|_{x_0}$$

Why? $\frac{df}{dx}$ reads as "a tiny change of f (df) divided by the tiny change of x (dx) that caused it". It is the **slope of the tangent** at x_0.



The Derivative: A Worked Example

Example 1: $f(x) = 2x$

$$\frac{f(x_0+h)-f(x_0)}{h} = \frac{2(x_0+h)-2x_0}{h} = \frac{2h}{h} = 2 \quad \Rightarrow \quad f'(x_0) = 2$$

Example 2: $f(x) = x^2$, using $a^2 - b^2 = (a+b)(a-b)$:

$$\frac{(x_0+h)^2 - x_0^2}{h} = \frac{(2x_0+h)\,h}{h} = 2x_0 + h \xrightarrow[h\to 0]{} 2x_0 \quad \Rightarrow \quad f'(x_0) = 2x_0$$

So $f'(5) = 2 \times 5 = 10$.

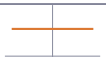
Derivative Function

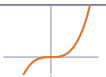
If f is differentiable at every point x, the function $f' : x \mapsto f'(x)$ is said to be the **derivative** of f.

e.g. for $f(x) = x^2$, the derivative function is $f'(x) = 2x$.

We have a **table of derivatives** for the classic functions, that we can use directly.

Table of Derivatives

$f(x)$	drawing	$f'(x)$
c (constant)		0
x		1
x^2		$2x$

$f(x)$	drawing	$f'(x)$
x^n		$n\,x^{n-1}$
e^x		e^x
$\ln x \quad (x > 0)$		$\dfrac{1}{x}$

Drawings are schematic (the e^x and $\ln x$ curves are rescaled to fit).

Using the Table

$$f(x) = x^3 \qquad \Rightarrow \qquad f'(x) = 3x^2 \qquad (x^n \text{ with } n = 3)$$

$$f(x) = x^5 \qquad \Rightarrow \qquad f'(x) = 5x^4$$

How Does a Computer Compute Derivatives?

We give it a function, for example:

```
def DontKnow(x):
    y = 3 * x**2 + x
    return y
```

How to compute `DontKnow'(x)`?

1. **Numerically** (finite differences): compute $\frac{\texttt{DontKnow}(x+h) - \texttt{DontKnow}(x)}{h}$ with a small h, e.g. $h = 10^{-5}$. Simple, but only approximate.
2. **Automatic differentiation**: the computer knows the derivative of each elementary operation ($+$, $\times$, `**`, `exp`…) and combines them with the rules below. Exact. This is what PyTorch does (`loss.backward()`).

Exercise (1/1)

Derivative by hand, then by computer. Let $f(x) = x^2$ and $x_0 = 3$.

1. Compute the right rate $\dfrac{f(3+h) - f(3)}{h}$ for $h = 1$, $h = 0.1$ and $h = 0.01$.
2. Compute the left rate $\dfrac{f(3) - f(3-h)}{h}$ for the same values of h.
3. What number do they approach? Compare with $f'(3)$ from the table.
4. Using the table, give the derivatives of x^4, e^x and $\ln x$.

Differentiation Rules and Properties

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Differentiation Rules

For f, g differentiable and $\lambda \in \mathbb{R}$:

- $(\lambda f)' = \lambda f'$
- $(f + g)' = f' + g'$
- $(fg)' = f'g + fg'$
- $(f \circ g)' = g' \times f' \circ g$ (the **chain rule**, see below)

Other notation for the last one: $\dfrac{df(g)}{dx} = \dfrac{dg}{dx}\dfrac{df(g)}{dg}$

Differentiation Rules: Examples

1. $f(x) = x^2 + x$ (sum)
   $f'(x) = 2x + 1$

2. $g(x) = x\,e^x$ (product, with $u = x$, $v = e^x$)
   $g'(x) = 1 \times e^x + x \times e^x = (1 + x)\,e^x$

3. $k(x) = e^{x^2}$ (composition, with $f = e^u$ and $g = x^2$)
   $k'(x) = g'(x) \times f'(g(x)) = 2x\,e^{x^2}$

Higher-Order Derivatives

- f'' is $(f')'$
- f''' is $(f'')'$
- $f^{(n)}$ is $\left(f^{(n-1)}\right)'$: differentiate n times

Example:

$$f(x) = x^2$$
$$f'(x) = 2x$$
$$f''(x) = 2$$
$$f'''(x) = 0$$

Searching for a Minimum / Extremum

Theorem 1

Let f be a differentiable function. If f has a local **extremum** (minimum or maximum) at x_0, then $f'(x_0) = 0$.

△**!** The converse is false: $f(x) = x^3$ has $f'(0) = 0$ but no extremum at 0.

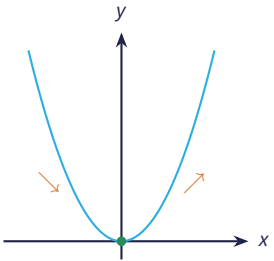
Theorem 2

- $f'(x) > 0$ on an interval means f is **increasing** there
- $f'(x) < 0$ on an interval means f is **decreasing** there

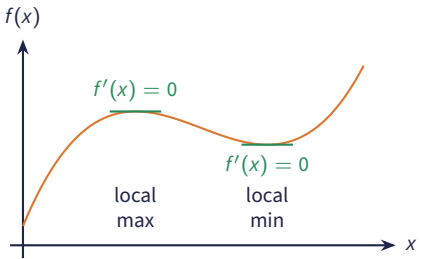
Searching for a Minimum: Example

$f(x) = x^2, \quad f'(x) = 2x$

- for $x < 0$: $f'(x) < 0$, f is decreasing
- for $x > 0$: $f'(x) > 0$, f is increasing
- $f'(0) = 0$: **minimum** at $x = 0$



Local vs Global Extrema

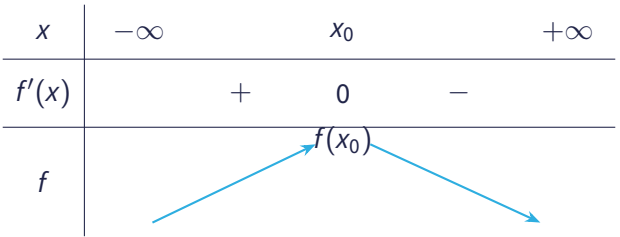


$f'(x) = 0$ at every local extremum (flat tangent).

A **local** extremum is the best point **in its neighbourhood**.
A **global** extremum is the best point **overall**.

Here the local max is not the global max: the curve keeps rising on the right.

Variation Tables



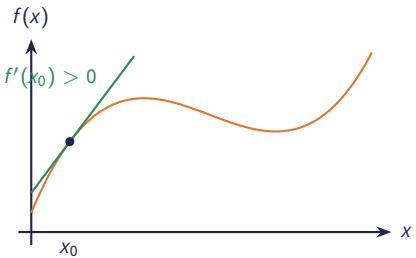
This allows to **find the extrema** (here a maximum at x_0). If there are several, you can then compare them: $f(x_0) \overset{?}{>} f(x_1)$, etc.

Exercise (1/1)

Let $f(x) = x^2 - x$.

1. Compute $f'(x)$.
2. Solve $f'(x) = 0$.
3. Draw the variation table of f.
4. What is the minimum of f, and where is it reached?

What If a Computer Wants to Find the Best Solution?



The computer only sees the function **locally**, around its current point x_0.

Locally, the curve looks like its **tangent**:

$$y = f'(x_0)\,(x - x_0) + f(x_0)$$

The sign of $f'(x_0)$ tells it which way is **down**.

Gradient Descent

General idea: we can move towards a local minimum by following the **gradient**, the vector composed of the derivatives: $\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right)$

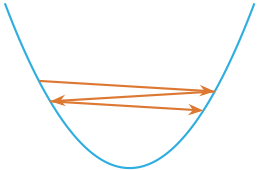
In 1D, $f : \mathbb{R} \to \mathbb{R}$: start at x_0 and do

$$x_{n+1} = x_n - \eta\, f'(x_n) \qquad (\eta > 0\text{: the } \textbf{step size}\text{, or learning rate})$$

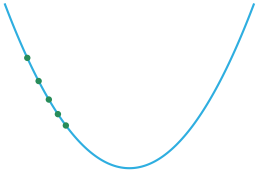
With a suitable η (see next slide), this converges to a **local** minimum.

$f'(x_n) > 0$ (going up to the right) $\Rightarrow$ we move left; $f'(x_n) < 0 \Rightarrow$ we move right.

Gradient Descent: Impact of η



large η: overshoots



small η: tiny steps

A too large η jumps over the minimum (and can even diverge).
But a too small η leads to a **large number of steps**.

The Chain Rule

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The Chain Rule

In a neural network, each **layer is a function**:

$$\text{output} = h_3 \circ h_2 \circ h_1\,(\text{input})$$

(h_1 is applied first, then h_2, then h_3.)

$\Rightarrow$ To train it, we need the derivative of a **composed** function.

The Chain Rule

The chain rule (2 functions)

$$(f \circ g)' = g' \times f' \circ g, \quad \text{i.e.} \quad (f(g(x)))' = g'(x) \times f'(g(x))$$

With this, we can compute the derivative of **complex** (composed) functions.

Other notation, same idea:

$$\frac{df(g)}{dx} = \frac{dg}{dx}\frac{df(g)}{dg}$$

Differentiate the outer layer, evaluated at the inner one, and multiply by the derivative of the inner one.

The Chain Rule: Example with 3 Functions

For three functions: $(f \circ g \circ h)'(x) = h'(x) \times g'(h(x)) \times f'(g(h(x)))$

Example: $k(x) = e^{(3x+1)^2}$, one layer at a time:

$$h(x) = 3x + 1, \qquad g(u) = u^2, \qquad f(v) = e^v$$

$$h'(x) = 3, \qquad g'(u) = 2u, \qquad f'(v) = e^v$$

$$k'(x) = \underbrace{3}_{h'} \times \underbrace{2(3x+1)}_{g'(h(x))} \times \underbrace{e^{(3x+1)^2}}_{f'(g(h(x)))} = 6(3x+1)\,e^{(3x+1)^2}$$

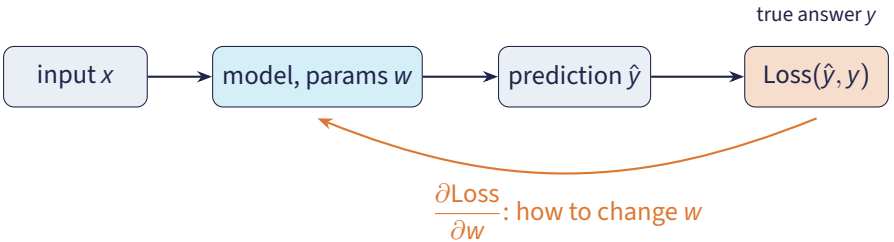
Backpropagation

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Why the Chain Rule Is Backpropagation

In ML: we evaluate our algorithm with a **loss**:

$$\text{Loss(parameters)} = \text{how good (or bad) the algorithm is}$$



Forward: compute the prediction and the loss. Backward: compute how each parameter changes the loss.

Why the Chain Rule Is Backpropagation

We want the loss to **decrease**. What can we do?

$$\frac{\partial \text{Loss}}{\partial \text{param}_1}, \quad \frac{\partial \text{Loss}}{\partial \text{param}_2}, \quad \dots$$

These tell us how to move parameter #1 or #2 to reduce the loss.

$\Rightarrow$ This is what **gradient descent** is about: $\text{param} \leftarrow \text{param} - \eta \frac{\partial \text{Loss}}{\partial \text{param}}$

$\Rightarrow$ Video reference: 3Blue1Brown, Neural Networks series (chapters on gradient descent and backpropagation)

Why the Chain Rule Is Backpropagation

Remark: differentiating **with respect to** a variable ("w.r.t.").

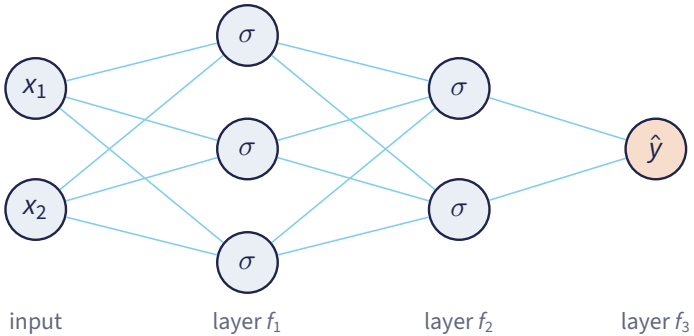
$k = u^2$ with $u = 3x + 1$:

$$\frac{\partial k}{\partial u} = 2u \qquad \frac{\partial k}{\partial x} = \frac{\partial u}{\partial x}\,\frac{\partial k}{\partial u} = 3 \times 2u = 6(3x + 1)$$

The same quantity k has a different derivative depending on what we differentiate it **w.r.t.**: the chain rule links the two.
(∂ is used instead of d when there are several variables.)

Neural Networks

Neural networks have **multiple layers**, with **activation functions**:



Each layer: a matrix product (Session 2) followed by an activation σ, e.g. ReLU.
So the output is a **composition** of the layers: $\hat{y} = f_3 \circ f_2 \circ f_1(x)$.

Neural Networks: Backpropagating the Loss

$f_{\text{tot}} = f_3 \circ f_2 \circ f_1$, and the loss compares the output with the solution:

$$\text{Loss} = (f_{\text{tot}}(x) - \text{sol})^2$$

So we need to **backpropagate** and get $\dfrac{\partial \text{Loss}}{\partial w}$ for **each parameter** w:

$$\frac{\partial \text{Loss}}{\partial w} = \frac{\partial f_{\text{tot}}}{\partial w} \frac{\partial \text{Loss}}{\partial f_{\text{tot}}} \qquad \text{with } \frac{\partial \text{Loss}}{\partial f_{\text{tot}}} = 2\,(f_{\text{tot}}(x) - \text{sol})$$

and $\dfrac{\partial f_{\text{tot}}}{\partial w}$ is itself computed with the chain rule, layer by layer, from the output back to the input.

Backpropagation

Backpropagation

Backpropagation is the fact of **recursively applying the chain rule** to a neural network, from the loss back to every parameter.

→ Video: 3Blue1Brown, “Backpropagation calculus”

Deliverable

Differentiate a composed function by hand, and write in one sentence how it connects to backpropagation.

Exercises

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Exercises (1/3)

Basic derivatives. Differentiate:

1. $f(x) = 3x^2 + 5x - 2$
2. $f(x) = x^4 - 2x$
3. $f(x) = 4e^x + \ln x$
4. $f(x) = x\,e^x$ (product rule)
5. $g(x) = e^{2x}$ (chain rule)
6. $h(x) = (3x + 1)^2$ (chain rule)

For question 6, check your result by expanding $(3x + 1)^2$ first.

Exercises (2/3)

Backprop in 2 layers. A tiny network with one neuron per layer:

$$h = w_1 x, \qquad \hat{y} = w_2 h, \qquad \text{Loss} = (\hat{y} - y)^2$$

with $x = 1, y = 2$ (the solution), $w_1 = 1, w_2 = 3$.

1. **Forward**: compute $h, \hat{y}$ and the Loss.
2. **Backward**: compute $\dfrac{\partial \text{Loss}}{\partial \hat{y}}$, then $\dfrac{\partial \text{Loss}}{\partial w_2} = h \dfrac{\partial \text{Loss}}{\partial \hat{y}}$ and $\dfrac{\partial \text{Loss}}{\partial h} = w_2 \dfrac{\partial \text{Loss}}{\partial \hat{y}}$.
3. Compute $\dfrac{\partial \text{Loss}}{\partial w_1} = x \dfrac{\partial \text{Loss}}{\partial h}$.
4. Do one gradient descent step with $\eta = 0.1$, then recompute the Loss. Did it decrease?

Exercises (3/3)

Activation layers.

$$\text{ReLU}(x) = \max(0, x) \qquad \sigma(x) = \frac{1}{1 + e^{-x}} \text{ (sigmoid)}$$

1. Compute $\text{ReLU}(-2)$, $\text{ReLU}(3)$. Draw ReLU.
2. What is $\text{ReLU}'(x)$ for $x < 0$? For $x > 0$?
3. Let $k(x) = \text{ReLU}(2x - 1)$. Using the chain rule, compute $k'(1)$ and $k'(0)$.
4. Compute $\sigma(0)$. Check that $\sigma'(x) = \sigma(x)\,(1 - \sigma(x))$, then compute $\sigma'(0)$.

Recap

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Recap

1. The derivative $f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h)-f(x_0)}{h}$ is the **instantaneous rate of change**: the slope of the tangent.
2. Table + rules: $(f+g)' = f' + g'$, $(fg)' = f'g + fg'$, $(x^n)' = nx^{n-1}$, $(e^x)' = e^x$, $(\ln x)' = 1/x$.
3. $f' = 0$ at an extremum; the sign of f' tells if f increases or decreases.
4. Gradient descent: $x_{n+1} = x_n - \eta f'(x_n)$ goes down towards a local minimum.
5. Chain rule $(f \circ g)' = g' \times f' \circ g$, applied layer by layer from the loss: **backpropagation**.

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To Do for Next Session

- Do the exercises
- Ask me questions :)